D (Printed Pages 4)

(20524) Roll No.

B.C.A.-II Sem.

18007

B.C.A. Examination, May-2024

DIGITAL ELECTRONICS AND COMPUTER ORGANISATION

(BCA-204)

Time : 3:00 Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** Sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt **all** the **five** questions. Each question carries **3** marks. Very short answer is required not exceeding **75** words. 5×3=15

1. NAND gate is known as universal gate. Realize the basic gates from this universal gate. 3
2. Differentiate between synchronous and asynchronous counter. 3

P.T.O.

3. Describe SR Latch? Draw its logic circuit and its truth table. 3
4. Differentiate between static memory, dynamic memory and auxiliary memory. 3
5. In computer memory, write down the different units for memory. A memory can store 1024 words each having 8 bits. Represent this memory in K (kilo). 3

Section-B

(Short Answer Type Questions)

Note : Attempt any **two** questions.

6. Minimize the following Boolean function using K-map: 7.5

   $F(A,B,C,D) = \Sigma(0,3,6,7,9,13,14,15)$

7. An AND gate has four inputs A,B,C,D and an output Y. 2.5×3=7.5
   (i) Write the Boolean expression and sequential diagram.
   (ii) When does the above gate will give high output? Draw the truth table.
   (iii) For $\overline{Y}$ write the Boolean expression and sequential diagram.

8. Describe Demultiplexer and design 1:4 demultiplexer using gates. 7.5

Section-C

(Descriptive Answer Type Questions)

Note : Attempt any **three** questions.

9. (a) Draw the logic circuit for the Boolean equation

$$Y = [A\overline{B}(C + BD) + \overline{AB}]C.$$ 7.5

(b) Simplify the expression and draw logic circuit for the simplified expression. 7.5

10. Draw the circuit of a serial in parallel out shift register. Draw and explain its working. 15

11. Draw the memory hierarchy in a digital computer. Explain each memory in terms of capacity, size, access time, implementation and managed by whom? 15

12. Draw the circuits of full adder and discuss its working. Draw its truth table also. 15

13. A memory has a capacity 32K X 8. Find. 3×5=15

(a) The numbers of words in memory.

(b) The numbers of bits in each word.

(c) The total numbers of bits in the memory.

(d) The total numbers of memory cells in the memory.

(e) Data input and data output lines.

A

(20623)

BCA – II Sem.

Printed Pages : 3

Roll No. .....

18007

B.C.A. Examination, June–2023

DIGITAL ELECTRONICS AND COMPUTER ORGANISATION

(BCA-204)

Time : 3 Hours] *[Maximum Marks : 75*

Note: Attempt questions from **all** Sections as per instructions.

Section–A

(Very Short Answer Type Questions)

Note : Attempt all the ***five*** questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 5×3=15

1. Give the logic diagram and truth table of RS flip-flop. 3

2. Why RAM is used in computer? How static RAM is different from Dynamic RAM and where these RAMs are used in computer system? 3

3. Construct half adder circuit using basic gates. 3

4. State and prove De-Morgan theorem. 3

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5. With logic diagram, characteristic table and the characteristic equation, explain the operation of a D Flip-Flop. 3

Section–B

(Short Answer Type Questions)

Note: Attempt any ***two*** questions out of the following three questions. Each question carries 7.5 marks. Short answer is required not exceeding 200 words. 2 × 7.5=15

6. What are multiplexer and Demultiplexer? Also give the logical expression. 7½

7. Draw and explain SISO, SIPO, PISO and PIPO shift registers with suitable example. 7½

8. Implement following boolean function using multiplexers. 7½

$F(ABCD) = \Sigma(0, 1, 3, 4, 9, 15)$

Section–C

(Descriptive Answer Type Questions)

Note: Attempt any ***three*** questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 3 ×15=45

9. Explain cache memory and cache initialization. Give the significance of cache memory.

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A two way set associative cache memory uses block of four words. The cache can accommodate a total of 2048 words from the main memory. The main memory size is 128K × 32

(a) Formulate all pertinent information required to construct the cache memory.

(b) What is the size of cache memory? 15

10. Design a synchronous sequential circuit with 2 inputs T and C. The output attains a value of 1 when T = 1 and C moves from 1 to 0. Otherwise output is 0. 15

11. What is decoder? Draw the circuit of a 2 to 4 decoder and explain its functions. 15

12. Explain the difference between all of the following- 15

(a) Volatile and Non-Volatile memory

(b) Static and Dynamic memory

(c) Sequential and Random Access memory

(d) Magnetic and Semi-Conductor memory

(e) Sum of Product (SOP) and Product of Sum (POS)

13. Design an 32 × 8 memory RAM chip using 16 × 4 memory chips. 15

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(20622) Roll No.

BCA-II Sem.

18007

B.C.A. Examination, June-2022

DIGITAL ELECTRONICS AND COMPUTER ORGANISATION

(BCA-204)

Time : Three Hours / *Maximum Marks : 75*

Note : Attempt **all** the Sections as per instructions.

Section-A

(Very Short Answer Questions)

Note : Attempt all **five** questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words.

1. Write Demorgan's Laws. 3
2. Differentiate between flip flop and latches. 3

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3. Differentiate between combinational logic circuit and sequential circuits. 3
4. Construct half substractor using logic gates. 3
5. Draw the memory hierarchy structure and mark the arrow from low to high (Speed) & high to low (Cost). 3

Section-B

(Short Answer Questions)

Note : Attempt any **two** questions.

6. Minimize the following Boolean function using K-map-. 7½

   $F(A,B,C,D,) = \sum (3,4,5,7,9,13,14,15)$

7. (i) Convert the SR flip flop to JK flip flop draw the truth table of JK flip flop also. 5

   (ii) Implement 4:1 multiplexer using 2:1 multiplexer 2½

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8. (i) simplify the expression: 4

$F(A,B,C,D)=ACD+\bar{A}B+\bar{D}$ by K-map.

(ii) How many flip flops are needed to implement a 32 bit register. 3½

Section-C

(Detailed Answer Questions)

Note : Attempt any **three** questions.

9. (i) Which gates are called universal gates and why? 5

(ii) Draw a full substractor circuit using NAND gate. 10

10. What do you mean by shift register? What is the need of shift register? Draw & explain bidirectional shift register. 15

11. Draw and explain 4-bit binary synchronous counter. 15

12. (i) Differentiate between EPROM & EEPROM. 5

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(ii) Differentiate between SRAM & DRAM. 5

(iii) Differentiate between L_1 cache & L_2 cache. 5

13. (i) Differentiate between ROM & PROM. 3

(ii) Describe USB. 3

(iii) Differentiate between primary memory and secondary memory. Also list the examples of primary memory and secondary memory. 3

(iv) Explain the concept of Virtual memory. 3

(v) Draw basic cell of memory. 3

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(21121) Roll No.

B.C.A.-IInd Sem.

18007 (CV-II)

B.C.A. Spl. and Back Paper Examination, Nov.-2021

DIGITAL ELECTRONICS AND COMPUTER ORGANISATION

(BCA-204)

Time : 1½ Hours] *[Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt any **two** questions. Each question carries **7.5** marks. Very short answer is required, not exceeding 75 words. 2×7.5=15

1. Prove that NOR and NAND gates are Universal gates.

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2. Explain the working of half-adder with suitable block diagram.

3. Draw the logic diagram of a Master slave D flip-flop. Using NAND gates.

4. What do you know about K-map?

5. Make the truth tables of :

   (i) half adder

   (ii) half subtractor

Section-B

(Short Answer Type Questions)

Note : Attempt any **one** question out of the following **three** questions. Each question carries **15** marks. Short answer is required not exceeding 200 words. 1×15=15

6. Prove that $(A+\bar{B}+AB)\ (A+\bar{B})\ (\bar{A}B)=0$

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7. Simplify the expression using Karnaugh map method.

$\bar{A}B\bar{C} - + AB\bar{C} + \bar{A}BC + ABC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$

8. Design a 3-bit binary down counter using S-R flip-flops.

Section-C

(Long Answer Type Questions)

Note : Attempt any **two** questions out of the following **five** questions. Each question carries **22.5** marks. Answer is required in detail. 2×22.5=45

9. Write short notes on following:

   (i) Virtual memory organization.

   (ii) Random access memories.

10. What do you mean by cache memory? How is the performance of a memory system improved by using cache?

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11. Design the binary counters having the following repeated binary sequence. Use JK flip-flops. https://www.ccsustudy.com

   (i) 0, 1, 2

   (ii) 0, 1, 2, 3, 4

12. (i) Implement a full adder with the help of two half adders and an OR gate.

   (ii) Design a Combinational circuit whose input is a four-bit number and whose output is the 2's Complement of the input number.

13. State De-Morgan's theorems of Boolean Algebra and prove them. What is the physical significance of these theorems?

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Total Questions : 13] [Printed Pages : 3

18007

B.C.A. IInd Semester Examination, May-2019

DIGITAL ELECTRONICS AND COMPUTER ORGANISATION

(BCA-204)

Time : 3 Hrs.] *[M.M. : 75*

Note :- Attempt all the Sections as per instructions.

Section–A

(Very Short Answer Type Questions)

Note :- Attempt all *five* questions. Each question carries 3 marks. Very short answer is required not exceeding **75** words.

1. Give the truth table and symbol for X-OR logic gate.
2. What is Subtractor ?

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3. Define associative memory.
4. Give the applications of shift registers.
5. State the boolean distributive law.

Section–B

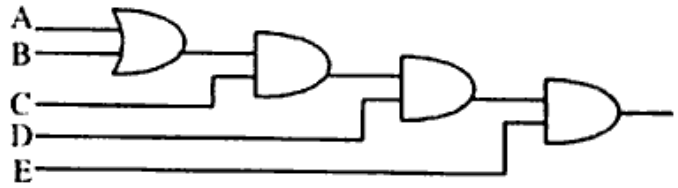
(Short Answer Type Questions)

Note :- Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding **200** words.

6. Simply the given function using *k*-maps :

   $F(A, B, C, D) = \Sigma(0, 2, 3, 5, 7, 9, 11, 13, 14)$

7. Derive the Boolean expression for logic circuit shown below :



8. Draw RS flip flop and explain its working.

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Section–C

(Long Answer Type Questions)

Note :– Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail.

9. Give difference between the following :

   (i) SRAM vs DRAM

   (ii) Registers vs Counters

10. Design 5-Mod counters using J-K flip flop.

11. (i) What is cache memory ? Why is it called high speed memory ?

    (ii) Design 8 × 1 Multiplexer

12. Describe the following terms :

    (i) ROM

    (ii) PROM

    (iii) EPROM

    (iv) RAM

    (v) Virtual Memory

13. Discuss various types of Logic Gates. Also discuss their applications.

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(20518) Roll No. .....

BCA-II Sem.

18007

B. C. A. Examination, May 2018

DIGITAL ELECTRONICS AND COMPUTER ORGANIZATION

(BCA-204)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very Short answer is required not exceeding 75 words. 3×5=15

1. What are universal gates ? Explain how basic gates can be realize using NAND and NOR gate.

2. Design and draw the logic diagram of full adder.

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3. What is Read Only Memory ? How PROM, EPROM, EEPROM differ from each other ?

4. What is flip-flop ? Explain T flip-flop.

5. Using K-map method simplify the following Boolean function :

$$F(ABCD) = \Sigma m(0, 2, 3, 6, 7) + \Sigma d(8, 10, 11, 15)$$

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words. 7½×2=15

6. What is race around condition ? Construct master slave flip-flop using SR flip-flop.

7. What is track and sector ? How data are stored in hard disc, floppy disc and CD ROM ? Explain.

8. Using eight 64×8 ROM chips with an enable input and decoder, construct a 512×8 ROM.

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Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. Design a synchronous sequential circuit with input A and B and output Y. Initially and at any time if both the inputs are 0, then the output Y is equal to 0. When A or B become 1, Y becomes 1 when other input also become 1, Y become 0. The output stay at 0 unit circuit goes back to initial state.

10. Discuss various semiconductor cells. Also discuss a RAM organization. If 16K×8 memory chips are used to construct 64K×16 memory :
    (a) Find how many chips will be needed
    (b) Draw block diagram showing connections of chips to address lines. https://www.ccsustudy.com

11. Explain and compare sequential and combinational circuit. Using full adder, design a four bit adder and subtractor circuit.

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12. Explain how 3 to 8 decoder function can be obtained from a demultiplexer.

13. Simplify the following Boolean function : 3×5
    (a) $W'X(Z'+YZ)+X(W+Y'Z)$
    (b) $X'Y+XY+XY'$
    (c) $XY'Z+X'Y'Z+XYZ$
    (d) $(X+Z')(Y+Z')$
    (e) $(A+D)(C'+D)(A+B'+C)$.

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(20517) Roll No.....

B.C.A. - II Sem.

18007

B.C.A. Examination, May 2017

Digital Electronics and Computer Organisation

(BCA-204)

(New)

Time : Three Hours] *Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

Section-A

Note : Attempt all **five** questions. Each question carries **three** marks.

1. What is truth table? What is its significance? 3
2. What is multiplexers? 3

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3. Explain the Edge Triggered D Flip-Flops. 3
4. Why are NAND and NOR gates more popular? 3
5. Difference between Registers and Counters. 3

Section-B

Note : Attempt any **two** questions.

6. Reduce the following Boolean expression using K-Map. 7.5

   $F(P, Q, R, S) = \Sigma(0, 3, 5, 6, 7, 11, 12, 15)$

7. The 2732 is a 4096 x 8 EPROM. How many address line does it have? 7.5
8. Draw the master slave JK flip-flop and explain its working. 7.5

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Section-C

Note : Attempt any **three** questions.

9. (a) Explain the difference between cache memory and virtual memory. 7.5

(b) Draw the Half adder Logic circuit and summarize the operation. 7.5

10. (a) State and verify De Morgan's Law in following Boolean Algebra. 7.5

(b) Draw a Logic Circuit Diagram for the Boolean expression 7.5

$$X : (Y'+Z)$$

11. (a) Explain the operation of the bi-directional shift register. 7.5

(b) Explain how a J-K flip-flop can be converted into a D flip-flop. 7.5

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12. Define the following : $3 \times 5 = 15$

(a) Multiplexer (8×1) MUX Design

(b) Register

(c) Flip-Flop Application

(d) Asynchronous Counter

(e) Basic Cell of Static RAM

13. Write short notes on cache memory organization. 15

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V

(20516) Roll No. .....

BCA-II Sem.

18007

B.C.A. Examination, May 2016

DIGITAL ELECTRONICS AND COMPUTER ORGANISATION

(BCA-204)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions of this Section. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 3×5=15

1. What are logic gates? Draw the schematic block diagram of a 3-input AND gate. Give its truth table. 3

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2. What is decoder and encoder? 3

3. What do you mean by Cache memory? 3

4. What is the function of shift-register? Where are such registers used? 3

5. Differentiate RAM and ROM. 3

Section-B

(Short Answer Questions)

This Section contains three questions, attempt any *two* questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words. 7½×2=15

6. Draw K-map and simplify the following boolean expression : 7½

$$Y(A,B,C,D) = \sum(0,2,4,6,8,10,12,14,15).$$

7. What is shortcoming of an S-R flip-flop? Explain how its shortcoming is removed by J-K flip-flop. Describe its operating principle. 7½

8. Explain the difference between static and dynamic memories. 7½

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Section-C

(Detailed Answer Questions)

This Section contains five questions, attempt any *three* questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. Explain the various Boolean laws and simplify the following boolean expressions : 5×3

   (a) $(\overline{A}+B+\overline{B}A)D$

   (b) $\overline{A}B\overline{C}+AB\overline{C}+\overline{A}\,\overline{B}\,\overline{C}+A\overline{B}\overline{C}$

   (c) $\overline{A}B\overline{C}+\overline{A}B\overline{C}D+B\overline{C}$.

10. (a) Implement a full-subtractor with two half-subtractors and an OR-gate. 7½

    (b) Show how a full-adder can be converted to a full-subtractor with the addition of one inverter circuit. 7½

11. Write short notes on the following : 3×5

    (a) EPROM

    (b) Dynamic RAM

    (c) Hard disk

    (d) Floppy disk

    (e) CD-ROM.

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12. Discuss the working principle of a synchronous (parallel) counter with its block diagram. What is the advantage of synchronous counters over asynchronous (ripple) counters? Give same example(s) of both types of counters. 15

13. What is associative memory? Draw and explain its block diagram. How read operations and write operations are performed in associative memories. 15

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